Investigating interactive affordances of agentic AI in creative ecosystems

PhD research proposal

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I would like to propose an in-depth study of the interactive affordances of agentic AI systems in creative domains and their application in designing tangible and embodied Human-AI interactions. The study will encompass an evaluation of existing tools and an exploration of how AI agents can be imagined and designed. The research will be situated in the domains of human-computer interaction and interaction design, and embodied cognition. The core methodology consists of longitudinal studies into the use of existing Human-AI co-creation systems, and a construction and evaluation of a novel open-source locally run agentic AI system for creative Human-AI collaboration for audio-visual performances. This proposal outlines gaps in the current research, possible design problems and suggestions for their resolution, as well as assessment strategies for this work.

1 BACKGROUND AND IMPETUS

My interest in creative tools comes from my experiences working on the intersection of embodied interaction and creative technology. After moving to the United States, I worked as a sound designer and VJ for Streb Dance company [1] touring and performing for 3 years. As a part of this collaboration, I developed an audio-visual system that captured the impacts of the bodies of the performers on various surfaces and translated them to audio effects and soundscapes that formed the musical accompaniment to the performance. This work was supported by a grant from the NY State Council on the Arts.

During my graduate studies at Parsons the New School for Design I became very interested in embodied and somatic interactions and generally in the human body as a means of artistic expression. I created a multimodal interactive tool that allowed visual artists to use a brain-computer interface and the movement of their body captured by a Kinect to create visual compositions [2]. This work continued as a series of collaborations on interactive public installations aimed at engaging users in creative play and exploring the locus of control in embodied interactions.

My most recent work was presented at the ACM IMX conference in 2025 and published in the proceedings [3]. The aim of this research was to engage with the opportunities and challenges presented by generative AI systems for embodied interaction in the context of visual narratives; this research was a collaboration with my graduate students at ArtCenter College of Design where I teach Creative Technology. My engagement with my students contributed significantly to my research into interactive affordances of creative tools and systems.

2 CONTEXT AND LITERATURE REVIEW

Creatives have engaged with algorithmic systems as collaborators, tools, and creative mediums for many years [65], and extensive research into the use of AI in creativity has followed this trajectory closely. Most of these studies present an uneven picture showing both advantages and disadvantages in involving generative AI in creative tasks [4, 6, 7]. Some studies examine the use of AI as a production assistant - simplifying laborious tasks and assisting in rapid exploration of possibilities [5], others point to true co-creation dynamics, and the applications of AI in selecting and synthesizing information, helping to accelerate decision-making and refine outcomes [7]. Grammenos and Lubart [57] assert that large language models (LLMs) score highly on various creative metrics in both producing and evaluating creative output, and highlight the fact that each of the three tested models exhibits a unique creative approach.

At the same time, many researchers acknowledge serious problems in creative uses of AI. Studies highlight the reduction in creative diversity [7, 58, 59, 61], "creative atrophy" due to overreliance on AI for cognitive processing [4, 60] and inhibition of the development of cognitive skills [7], lack of humor [57], novelty or surprise [62], and warn that AI often produces biased or even intentionally malevolent output [7, 61].

The most valuable research in the context of this proposal focuses on specific tasks and processes, like critical reasoning [4] or ideation [5], or on specific mediums, like screenwriting [9, 10], but most of these studies investigate generative rather than agentic AI. I believe that the coming of age of agentic AI systems drives a radical shift in creative engagement with AI and offers exciting opportunities to develop new interactive paradigms in this domain. The investigation of these opportunities is the core of this proposal.

2.1 Human-AI Co-Creation frameworks

In recent years human-AI co-creativity was called a "grand challenge for HCI" [66] and significant effort was dedicated to the development of comprehensive conceptual frameworks for understanding Human-AI co-creativity [42, 43, 44, 49, 58, 63]. Of particular interest to this proposal is the conception of the human-computer co-creation within the enactive cognitive model [45, 46], part of the embodied, embedded, extended, and enacted (4E) philosophy of mind [75]. This framing points to strong connections between tangible and embodied interaction [76], and human-AI co-creativity, inviting exploration of material communication [77] between human and algorithmic co-creators. It also shifts the focus to the in-situ exploration and playful engagement over consistency, reliability and predictability emphasized by frameworks like UTAUT and psychological ownership theory [74] or Human–AI Co-Creative Design Process (HAI-CDP) model [83].

Frameworks for building and evaluating co-creation systems based on these theoretical foundations have recently been developed. The Co-Creative Framework for Interaction design (COFI) [47] outlines a set of design principles stressing the importance of communication and collaboration styles. The Co-Creative Design Framework for Hybrid Intelligence [48] enhances this approach by incorporating principles from enactive and embodied cognitive science. I envision using this conceptual foundation for the proposed research to better understand constraints and affordances of embodied interaction, and modes of creative engagement with AI.

I am particularly interested in exploring two modes of creative engagement outlined in these frameworks: improvisational [49] and divergent [44, 47], since the more extensive exploration of these modes was specifically called for by the current researchers [63]. I would propose an alternative framing of the improvisational mode as "resonant" [80], in a sense that under the right conditions it produces mutually amplifying signals, in response to the recent calls for the AI to be regarded as complementary amplifiers of the human cognition [58]. This type of engagement has already been explored in the context of design for AI and robotic interactions [22], and I propose extending this research into the design of creative agentic systems.

The divergent mode is often understood as a wide and overarching term describing several variations of cognitive engagement [84, 85]. I am primarily interested in "adversarial" or "provocateur" mode [86] where AI would be challenging and provoking users to introduce creative friction [81, 82]. This type of engagement has been explored as a mode of human-human [23] and human-robot [24] creative collaborations, but in the field of Human-AI interaction the AI models remain perpetual “yes men”, and the absence of friction and pushback between users and AI models has been viewed as problematic [25, 55].

Researchers have called for efforts to examine the performance on both divergent and convergent creative tasks, as well as the use of AI tools to support processes of creative metacognition and self-regulation of creative action [63]. I hope to respond to this call in this proposed study. Additionally, a recent review of Human-AI co-creation systems identified several open challenges in this space [49]. As a part of this proposal I would like to focus on the issues of

- “Responding Adaptively”: the AI systems respond to changes in user’s creative behavior [53],
- “Dynamic Meaning Construction”: creative output emerges as a product of interactions between a human and an AI system [45],
- “Improvisational Creativity”: An AI system should have the ability to understand unexpected input from the user, build upon in, and be able to produce unexpected output, thus stimulating uncertainty as a creative force [54],
- Open-Ended”-ness: Rather than pursuing pre-defined goals the AI system should both stimulate and incorporate emergent goals [42].

2.2 Tangible and embodied interactions

I believe that tangible and embodied interaction models hold significant promise in the research of human-AI interactions [15, 16]. The use of TEI to enhance users’ cognitive and creative abilities is well researched and documented [69, 70, 71]. Recent studies explored the application of this domain creative reflection [17], tactility [18] and gestural interaction [19] for musical performance and production, as well as specific framing of human-AI interactions for creative engagement [20] and intelligibility [21]. Studies have shown that generative AI systems that map tangible inputs to creative outputs allow users to engage in exploratory, trial-and-error behavior and enhance their feeling of control [67]. A compelling case has also been made for the design that allows computer systems, in particular robots, to generate output for the users employing movement and motion [72].

This is exciting research, but it remains sparse, and calls have been made for additional exploration in this area [68]. I believe that significant contributions can be made at the intersection of tangible and embodied interactions and agentic AI and would like to address this in this proposed study.

2.3 Agentic AI in creative systems

The key distinction between generative and agentic AI systems lies in the mode of user engagement - while generative AI models require prompting to produce output, agentic systems are capable of more autonomous operations with much more loosely defined goals. This means that agentic systems employ internal reasoning and goal-setting structures to be able to act with a substantial degree of independence [8].

Only very recently have studies begun to identify the possibilities for creative engagement with agentic systems [13] and laid out frameworks for designing and evaluating tools for this engagement [14, 85, 91]. The study of the application of agentic AI in creative tasks remains both limited in duration of study and usually rooted in a specific medium like drawing [50], musical performance [90], interaction design [11] or worldbuilding [12].

This seems to be a significant gap: a longitudinal study could reveal how co-creation with agentic systems unfolds over an extended period. Calls have been made for longitudinal studies focused on the creativity rather than usability of creative tools [66] and to examine how the sense of ownership evolves in long-term creative practices [73]. I believe it would be useful to supplement these concerns with a study of interactive affordance of specifically agentic AI systems that could be used to form a set of design principles. This proposal seeks to address these concerns.

3 RESEARCH QUESTIONS

The core question I proposed to engage with during this research is:

> What are the interactive affordances of a materially anchored agentic AI system that could be leveraged to amplify improvisational and divergent modes of human creativity within a long-term creative collaboration with an agentic AI system?

There are closely entangled sub-questions that I believe would be useful to investigate in connection with the main question:

1. How can these affordances be applied to design tangible and embodied interfaces for adaptive, responsive and improvisational human-AI co-creation systems for audio-visual performance?
2. How can human-AI co-creative systems be designed to contribute to dynamic meaning construction and aesthetic human agency during the creative collaboration with autonomous AI agents?

I would like to conduct this research on the intersection of the following domains:

- Human-Computer Interaction. This would be the core domain of inquiry, drawing on the established methodologies and frameworks of this discipline.
    - Tangible and Embodied Interaction. This proposed research will rely on the established methods in this subdomain and continue to explore the new ones.
    - Interaction Design. I am hoping to engage in design through research and participatory design methodologies, as well as established practices for interface design for creative engagement.
- Embodied cognition. As a subdomain of cognitive science dealing with the use of the human body and its interactions with the environment, this field of inquiry would be useful in providing evaluation frameworks and insights into modes of creative engagement.

4 METHODOLOGY

4.1 Evaluating existing systems

There is a collection of useful precedents that I can rely on to build the foundation of my proposed research. Music systems from GenJam [92] to Neural Notes [26] and Automatar [27], a long list of generative image models, most notably Abraham [28], and world-building AI collaborators like Emergentic [12, 29] all provide useful insights into the design, construction and the modes of interactive and creative engagement with AI systems.

In order to examine the proposed research questions, I hope to carry out the evaluation of two existing systems for creative human-AI engagement: Tangible Co-Ideation Across Time [30] and jam_bot [31]. The former proposes engagement with AI assistants through tangible avatars and fine-tunes large-language models from a personality of a human to assist in ideation and concept development. The latter, jam_bot, is an AI assistant capable of real-time music generation for the purpose of collective improvisation and exploration of musical themes.

While neither would be considered an agentic system because regular prompting in one form or another is still necessary for the intended functionality, these two models cover a wide variety of functional and interactive paradigms, and I believe will serve as a useful springboard for further exploration.

Tangible Co-Ideation Across Time could provide insights into the dynamic meaning construction during the exploratory and ideational phases of collaborative human-ai co-creation, particularly how the material anchoring of human-ai communication through a tangible interface affects the development of trust in the AI partner [87] and a sense of agency for a human creator [88, 89]. Jam-bot situated on the border between an instrument and a band mate could be used to evaluate how a responsive and adaptive system can amplify improvisational creativity in a long-term creative relationship.

I propose both of these studies as longitudinal studies with expert users. Researchers have highlighted the lack of expert participants and theoretical grounding in existing studies of human-AI co-creativity [66], and I am hoping to fill this gap before moving on to the examination and design of the specifically agentic systems.

4.2 Designing and building a system for creative engagement with agentic AI

The evaluation of these systems will serve as a foundation to design and build a Human-AI creative ecosystem for tangible and embodied interaction with an agentic AI for audio-visual performances. This system will constitute research through design and making, and will draw on the co-creative design framework for hybrid intelligence [48] and material engagement framework [32] in its design approach. It will allow me to further extend the inquiry into the research questions, particularly the investigation of creative friction and divergent creativity in human-ai collaboration, and the role of tangible and embodied interfaces in amplifying resonant co-creativity. It will also allow me to compare agentic and non-agentic AI tools and further examine how an autonomous agentic AI system can be constructed to be responsive, adaptive and divergent in its creative approach.

I would also address the call issued by Téo Sanchez at al. [65] to design a system that ensures interoperability of the AI and non-AI based components to allow for modularity and combination of computational methods.

In proposing this system I am making several assumptions that I am hoping to evaluate during this research:

- It is possible to construct a multimodal agentic AI system [51, 52] capable of serving as a creative partner for an artist that would run locally on a personal computer,
- An agentic AI system can promote creative co-evolution of a human artist and agentic AI system leading to more intuitive, enjoyable, and supportive of human creative expression [56],
- Intelligible co-presence of a human artist and agentic AI system as understood and designed through the perspective of embodied cognition will foster “creative resonance” or “creative flow [79],”
- The shift of focus of interface design from the outcome to the process of creative engagement will ensure constructive creative relationship.

4.3 Evaluation frameworks

To evaluate these assumptions and the results of the proposed study in general I propose a longitudinal study on how engagement with semi-autonomous agentic AI systems affects human creativity.

This evaluation will measure the following components:

- Human Experience: evaluating how the AI agent alters the users’ creative flow [79] and cognitive load measured through Creativity Support Index [33] and assessing possible task execution issues via NASA Task Load Index [34].

- Creative impact of the designed novel Human-AI interaction system can be assessed with a range of techniques, in particular Consensual Assessment Technique [35].
- The quality of the developing Human-AI partnership can be evaluated by measuring qualitative parameters like sense of agency, creative agency, and the locus of control using custom Likert scales [36].
- Reconceptualization of creativity as a distributed property of a human-AI collaboration could open possibilities for the new ways to measure it [78].

This is, of course, only the preliminary proposal for the evaluation methodology, and the trajectory of the design and the research efforts will surely suggest more suitable options.

4.4 Technical requirements, challenges, and possibilities

There are also several technical challenges that will be necessary to overcome during this research:

1. The proposed design and construction of a creative human-AI ecosystem would be based on open-source and locally run software components. I believe that this to be an important component of the study for the reasons of both technical control and ethical engagement with agentic AI. This might present limitations on the types of models that would be possible to run on typical modern consumer-level computers. Currently, two approaches dominate the technical landscape, each with its own drawbacks and advantages. Dedicated GPU, typically one from NVIDIA, with a significant amount of dedicated RAM is capable of very fast performance with smaller models but the performance quickly slopes off with the larger ones. Apple Mac line of computers offers a unified memory approach that allows them to share CPU RAM for the use of the GPU, extending acceptable performance well into the larger models that offer better overall results [37, 38]. I believe these limitations can be resolved by optimizing the performance of a dedicated local system for specific tasks. Also, the trajectory of technical development points to the significant improvement of the performance of smaller models in the coming years making them more suitable for use in the proposed system.
2. Several researchers pointed out the lack of creative friction in large language models as a problem for creative engagement [25, 7, 4]. The lack of unexpected or random outcomes so productive for creative tasks might be the outcome of the fundamental structure of the stochastic language and reasoning models as they produce the most probable rather than the most “interesting” results [39]. I believe this issue could be overcome by continuous fine-tuning of the underlying model in the process of creative engagement. This points to the possibility of fostering a creative partnership with AI but at the same time might raise the barrier to entry and require significant investment on the part of the artist. I believe that overcoming this challenge will be one of the most significant contributions of this proposed study.
3. Persona drift has been identified as one of the major issues with AI assistants and agentic AI [5, 40]. This might present a challenge in the development of a system meant to act as a creative partner. Recent research has made significant progress in overcoming this issue [41] and allows hope of even more robust solutions in the future, but a significant amount of work is needed to ensure stability. Possible silver lining is that it might point to a potential path to overcoming the derivative and predictable behavior of a model. I believe that I would be able to contribute to this research, and that this contribution would constitute another important outcome of this proposed research.

5 CONCLUSION

This document outlines my proposal for a doctoral study of the interactive affordances of agentic AI in creative ecosystems. I propose evaluating 2 existing systems, and creating and studying a new one. I believe that a longitudinal study of the effects of interactions between human artists and agentic AI will help us better understand both the possibilities for human creative engagement with AI and the opportunities for constructing tangible and embodied interfaces for Human-AI interactions.

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